DFA Minimizer

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The following is an auto-explainer for DFA Minimization as represented by a series of figures depicting the evolving partitions of the DFAâ€™s state set.

A Deterministic Finite Automaton (DFA) is denoted by the quintuple

$$M = (Q, \Sigma, \delta, q_0, F)$$

and the minimization process works by iteratively refining partitions of the state set until all equivalent states are grouped together. In the minimization process:

- The initial partition divides Q into two blocks: the final states F and the non-final states $Q \setminus F$.
- At each subsequent step, the partition is refined by examining the transitions for every input symbol and grouping together states that behave equivalently.
- This iterative refinement continues until no further splitting is possible, thus yielding a minimal DFA.

1 DFA Definition and Minimization Overview

A Deterministic Finite Automaton is defined as a 5-tuple

$$(Q, \Sigma, \delta, q_0, F)$$

where:

- Q is a finite set of states,
- Σ is a finite input alphabet,
- $\delta : Q \times \Sigma \to Q$ is the transition function,
- $q_0 \in Q$ is the start state, and
- $F \subseteq Q$ is the set of final (accepting) states.

The minimization of a DFA entails computing an equivalent DFA with the fewest possible states. The process begins by partitioning the state set into final and non-final states and then repeatedly refining these partitions by identifying states that are indistinguishable with respect to the input symbols. The resulting minimized DFA retains the same language as the original.

2 DFA Transition Function

The DFA's transition function, δ, is defined as follows:

$$\delta(1,\, a) = 2$$
$$\delta(1,\, b) = 4$$

$$\delta(2,\, a) = 4$$
$$\delta(2,\, b) = 3$$

$$\delta(3,\, a) = 3$$
$$\delta(3,\, b) = 3$$

$$\delta(4,\, a) = 2$$
$$\delta(4,\, b) = 5$$

$$\delta(5,\, a) = 6$$
$$\delta(5,\, b) = 5$$

$$\delta(6,\, a) = 4$$
$$\delta(6,\, b) = Z$$

$$\delta(Z,\, a) = Z$$
$$\delta(Z,\, b) = Z$$

3 Indistinguishability Table Steps

We apply the table-filling (indistinguishability) algorithm iteratively. For each iteration, we show the triangular table and provide a short explanation of which pairs were marked.

Table Step 1

Explanation: Initially marked pairs where one state is final and the other non-final: 1,3, 2,3, 3,4, 3,5, 3,6, 3,Z

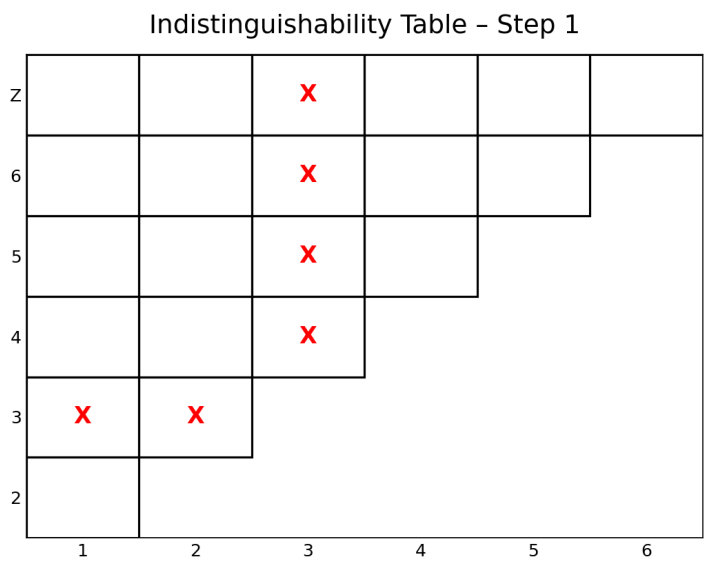


Table Step 2

Explanation: Newly marked pairs: 1,2, 2,4, 2,5, 2,6, 2,Z

Indistinguishability Table – Step 2

	1	2	3	4	5	6
Z		X	X			
6		X	X			
5		X	X			
4		X	X			
3	X	X				
2	X					

Table Step 3

Explanation: Newly marked pairs: 1,5, 1,6, 1,Z, 4,5, 4,6, 4,Z

Indistinguishability Table – Step 3

	1	2	3	4	5	6
Z	X	X	X	X		
6	X	X	X	X		
5	X	X	X	X		
4		X	X			
3	X	X				
2	X					

Table Step 4

Explanation: Newly marked pairs: 1,4, 5,6, 6,Z

Indistinguishability Table – Step 4

	1	2	3	4	5	6
Z	X	X	X	X		X
6	X	X	X	X	X	
5	X	X	X	X		
4	X	X	X			
3	X	X				
2	X					

Table Step 5

Explanation: Newly marked pairs: 5,Z

Indistinguishability Table – Step 5

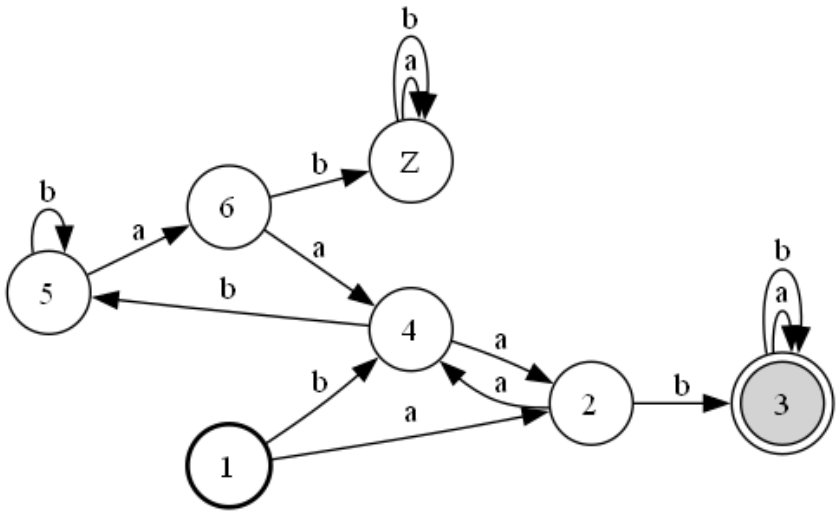
	1	2	3	4	5	6
Z	X	X	X	X	X	X
6	X	X	X	X	X	
5	X	X	X	X		
4	X	X	X			
3	X	X				
2	X					

Table Step 6

Explanation: No new pairs marked; table is now stable.

Indistinguishability Table – Step 6

	1	2	3	4	5	6
Z	X	X	X	X	X	X
6	X	X	X	X	X	
5	X	X	X	X		
4	X	X	X			
3	X	X				
2	X					

4 Minimized DFA

After no further distinguishable pairs can be marked, the remaining unmarked state-pairs are equivalent and are merged. The resulting minimized DFA is: